

Target Region: CENTRAL INDIA	Daily Volume: 105 GB/day	BI Users: 85 Users	Workload Mix: BI_ENG
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Executive Architecture Recommendation

Enterprise Platform Decision & Sizing Saturated Matrix

Data Confidence:  98%

RECOMMENDED PLATFORM

Azure Databricks

87% MATCH

Based on your input of 105 GB/day processed, 85 peak BI users, and mixed team skillset, Azure Databricks is the recommended architecture platform for your enterprise workload.

\$ COST / MO

₹1,78,500 – ₹2,76,250

SKU SIZING

8-Node SQL Warehouse + 1-Node Jobs Cluster

CONFIDENCE

98%

MIGRATION

Medium

DURATION

8–12 Weeks

POTENTIAL SAVINGS

18% - 35%

SCANNABLE DECISION BADGES

✓ Superior Spark Engine

✓ Industry MLflow MLOps

✓ Unity Catalog Governance

✓ Granular Spot VM Scaling

⚠ Dual-Billing Complexity

⚠ Requires Cluster Admin

⚠ No Native Direct Lake Zero-Copy

✓ WHY AZURE DATABRICKS WAS SELECTED:

- For your bi + eng workload, Azure Databricks provides the optimal balance of cost, operational simplicity, and feature velocity.
- Existing team skills and target SLA window fit comfortably within default capacity allocations.

⚠ WHY NON-SELECTED PLATFORM WAS NOT CHOSEN:

- The non-selected platform carries higher projected TCO (₹9,56,250 – ₹19,08,250) for this workload mix.

- Requires additional architectural setup and operational overhead to achieve equivalent performance.

 Migration Roadmap & Timeline Estimate

Enterprise Transition Assessment

MIGRATION DIFFICULTY

Medium Complexity



ESTIMATED DURATION

8–12 Weeks

~320 Total Engineering Hours

RISK LEVEL

 **Low Risk**

Standard automated ingestion & schema migration



Microsoft Fabric

Fit Score: 85/100

Unified SaaS Analytics Platform

Data Confidence: 98%

₹9,56,250 – ₹19,08,250

Estimated Monthly Cost Range

₹1,14,75,000 – ₹2,28,99,000 / yr

\$ Cost Itemization Breakdown

Monthly

Capacity Compute (CU)	₹7,82,000
OneLake Storage	₹2,991
SQL Data Warehouse	₹1,70,000
Power BI Integration	Included in F-SKU

 Suggested Sizing: F64

PRIMARY SIZING DRIVER

Peak concurrent BI users (85 users → ~44 CUs floor)

Key Sizing Assumptions:

- Fabric Capacity pay-as-you-go rate in Central India (Pune) (~₹9,51,976/mo base for F64).
- 730 hours/month capacity availability with dynamic smoothing and bursting enabled.
- OneLake storage estimated at 1530 GB (₹2,991/mo).



Azure Databricks

Fit Score: 87/100

Managed Spark & Lakehouse Engine

Data Confidence: 98%

₹1,78,500 – ₹2,76,250

Estimated Monthly Cost Range

₹21,42,000 – ₹33,15,000 / yr

\$ Cost Itemization Breakdown

Monthly

Databricks Compute (DBUs)	₹1,36,000
ADLS Gen2 Storage	₹2,601
Azure VM Infrastructure	₹72,250
Unity Catalog Governance	Unity Catalog Included

 Suggested Sizing: 8-Node SQL Warehouse + 1-Node Jobs Cluster

PRIMARY SIZING DRIVER

Jobs Compute clusters (1 nodes) for scheduled batch ETL

Key Sizing Assumptions:

- Databricks DBU rates in Central India (Pune) with applied regional multiplier.
- EXCLUDES Azure VM Infrastructure costs (VM instance pricing billed separately by Azure).
- ADLS Gen2 storage estimated at 1530 GB (₹2,601/mo).
- SQL Warehouse auto-stop configured at 10-15 minutes of inactivity.



Side-by-Side Architectural Matrix

Fabric Score: 85/100

Databricks Score: 87/100

ARCHITECTURAL DIMENSION	MICROSOFT FABRIC	AZURE DATABRICKS	ADVANTAGE
Cost & Licensing	Optimal ₹9,56,250 – ₹19,08,250/mo (Unified F-SKU)	Moderate ₹1,78,500 – ₹2,76,250/mo (DBUs + VM)	DATABRICKS
Scalability	Moderate Auto-smoothing CU capacity (F2 to F512)	Optimal Extreme horizontal cluster autoscaling & multi-cloud	DATABRICKS
Power BI Integration	Optimal Native Direct Lake zero-copy reporting	Complex SQL Warehouse connector (import/DirectQuery)	FABRIC
Data Engineering	Moderate Data Factory Pipelines & Synapse Notebooks	Optimal Delta Live Tables (DLT) & PySpark native	DATABRICKS
Machine Learning	Complex Synapse ML & AutoML (standard)	Optimal Native MLflow, Feature Store, GPU clusters	DATABRICKS
Streaming Processing	Moderate Eventstream & Real-Time Intelligence	Optimal Structured Streaming & Delta Engine	TIE
Data Governance	Optimal Microsoft Purview & Workspace roles	Optimal Unity Catalog (Cross-cloud fine-grained)	DATABRICKS
Operational Complexity	Optimal Low (Fully Managed SaaS)	Complex Medium-High (Cluster & Workspace Admin)	FABRIC
Team Learning Curve	Moderate Low	Optimal Minimal	DATABRICKS
Time to Production	Optimal 2–4 weeks (Turnkey SaaS workspace)	Moderate 4–8 weeks (Workspace & cluster setup)	FABRIC



Ask AI Architecture Copilot

Live OpenRouter AI

Interactive Architectural Advisory · OpenRouter AI Engine

SUGGESTED ENTERPRISE QUESTIONS (1-CLICK ASK AI):

Cost Optimization & Best Practices

MICROSOFT FABRIC

Save up to ₹3,65,500/month (~38%)

Switch from Pay-As-You-Go to 1-Year Capacity Reservation

Commit to a 1-year Fabric Capacity reservation for baseline F-SKU workloads.

Actionable Steps:

- ✓ Evaluate 30-day peak Capacity Unit (CU) utilization in Fabric Capacity Metrics app.
- ✓ Purchase 1-year reserved capacity for baseline F-SKU in Azure Portal.
- ✓ Keep pay-as-you-go enabled for seasonal burst capacity.

AZURE DATABRICKS

Save up to ₹63,750/month (~35%)

Configure Auto-Stop & Utilize Spot VM Instances for Batch Jobs

Set aggressive auto-stop timeouts on SQL Warehouses and configure Spot VMs for non-critical pipelines.

Actionable Steps:

- ✓ Set SQL Warehouse auto-stop to 10 minutes of inactivity.
- ✓ Enable Spot VMs with fallback to On-Demand for daily ETL Jobs Clusters.
- ✓ Migrate ad-hoc analysis notebooks to Serverless Compute to pay strictly for query duration.

BOTH

Reduced memory overhead & zero data movement

Adopt Direct Lake & Delta Parquet Storage Standards

Store all silver/gold data in open Delta Lake format to eliminate ingestion hops and import mode refreshes.

Actionable Steps:

- ✓ Standardize all data transformations on Delta Parquet format.
- ✓ Use OneLake shortcuts or Unity Catalog external locations for zero-copy access.
- ✓ Enable Direct Lake mode on Power BI semantic models for instant report performance.

Estimated using publicly available Microsoft and Databricks list pricing as of July 2025. All cost figures are directional estimates for **Central India (Pune)**. Actual costs depend on region, workload characteristics, infrastructure choices, autoscaling, reserved pricing, enterprise agreements, and future pricing updates. **Verify estimates before making purchasing decisions.**

India Pricing Note: Fabric Central/South/West India rate confirmed at \$0.24/CU-hr vs \$0.18 in US East (~33% premium). Databricks DBU rates are region-stable, but underlying Azure Central India VMs run ~15-25% above US East.